

DRY-RUN GUIDE | digonal.py

Source: Python\3rd_Belt\Matrix\digonal.py

CORE BEHAVIOR

Sums the main-diagonal entries of a square matrix.

REPRESENTATIVE INPUT

3x3 rows '1 2 3', '4 5 6', '7 8 9'.

STEP TRACE

Take [0][0]=1, [1][1]=5, [2][2]=9; sum=15.

EXPECTED OUTPUT / RETURN

'15'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\digonal.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Find the trace of a square matrix by summing its main diagonal elements.
```

```
002 | # Add your solution here.
```

```
003 |
```

```
004 | def trace_of_matrix(matrix):
```

```
005 |     return sum(matrix[i][i] for i in range(len(matrix)))
```

```
006 |
```

```
007 | if __name__ == "__main__":
```

```
008 |     n = int(input())
```

```
009 |     matrix = [list(map(int, input().split())) for _ in range(n)]
```

```
010 |     print(trace_of_matrix(matrix))
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.